

Proof for 17-Gon and 257-Gon Line-Only Challenges Being Impossible

Setup

The supplied parser begins with exactly two points,

$$O = (0, 0) \quad \text{and} \quad P = (1, 0),$$

and no predefined lines or circles. The remote verifier requires one initial circle and then enforces the line-only restriction:

`This challenge only allows lines (after the initial circle) - no circle commands permitted`

Since only O and P exist at the start, the initial non-degenerate circle must be centered at one of them and pass through the other. Constructing the circle creates a circle object, but it creates no new points.

Let ℓ be the line through O and P . In the parser's coordinate system, ℓ is the x -axis.

The Common Obstruction

Theorem 1. *Under this initialization and line-only restriction, no non-degenerate regular n -gon can be constructed for any $n \geq 3$.*

Proof. We prove the following invariant:

Every constructed point lies on ℓ , and every constructed line is geometrically equal to ℓ .

The invariant is true initially because the only points are O and P , both of which lie on ℓ . The required initial circle does not add a point, so the invariant is still true after the first command.

Assume that the invariant holds after some number of commands. A new line must be drawn through two existing points. Both points lie on ℓ , so the new line is again ℓ . Intersecting two constructed lines cannot produce a point because both lines are the same line and therefore have no unique intersection. This matches the supplied parser, whose line-line intersection routine rejects a zero determinant.

The result is unchanged even if the line-only rule is interpreted generously enough to allow intersections with the single initial circle. Every constructed line is ℓ , so every point obtained by intersecting such a line with the circle also lies on ℓ . Any later line through those points is still ℓ . There is only one circle, and intersecting it with itself is rejected as a concentric circle-circle intersection.

Thus every available operation preserves the invariant. By induction, every point obtainable by any finite construction lies on ℓ .

The final `n_gon` command does not create points; it only checks the listed vertices. A non-degenerate regular polygon requires at least three non-collinear vertices. The parser checks this condition in its concyclicity test and rejects a polygon when its first three vertices are collinear. Since all obtainable points lie on ℓ , every proposed regular n -gon with $n \geq 3$ must fail this test. $\square$

The Two Challenges

Corollary 1 (17-gon). *No valid regular 17-gon can be constructed under the line-only rules.*

Proof. Apply the theorem with $n = 17$. The final command requires 17 distinct vertices, beginning with P , but every obtainable vertex lies on ℓ . Therefore the vertices cannot form a non-degenerate regular 17-gon. $\square$

Corollary 2 (257-gon). *No valid regular 257-gon can be constructed under the line-only rules.*

Proof. Apply the theorem with $n = 257$. The final command requires 257 distinct vertices, beginning with P , but every obtainable vertex lies on ℓ . Therefore the vertices cannot form a non-degenerate regular 257-gon. $\square$

Why No Difference in Proof?

The two shapes differ in size, but not in the reason they fail:

	17-gon	257-gon
Required vertices	17	257
Additional vertices after P	16	256
Fermat-prime form	$2^{2^2} + 1$	$2^{2^3} + 1$
Classically constructible	Yes	Yes
Constructible here	No	No

Both polygons are classically constructible with an unrestricted straightedge and compass. That fact does not help in this challenge, because the restricted construction never produces a third non-collinear point. The 257-gon requires more final vertices than the 17-gon, but neither construction can begin under the current line-only contract.

Conclusion

The 17-gon and 257-gon challenges do not require two different impossibility arguments. They are two instances of the same degeneracy: starting from only O and P , one initial circle followed by line-only operations leaves every constructible point on OP . Since no non-degenerate polygon can have all of its vertices on one line, both targets are impossible under the stated rules.

Small Note

If anyone is wondering why the 17-gon line-only and 257-gon line-only categories previously had accepted submissions, it is because the parser contained a bug that could be exploited to make invalid constructions pass the verification system. This has been confirmed by the UIUCTF hosts. Those submissions therefore do not contradict the result of the proof that, according to the original rules, no genuine valid submission exists.